Functional Requirement Document / Technical Specification Document

Project Name: **HealthConnect Data Intelligence**: Sept, 2025

Revision Sheet

Release No.	Date	Revision Description
1	Sep-2025	First Version of HealthConnect Data Intelligence: Sept, 2025

1.0 Purpose

The purpose of this document is to define the functional and technical requirements for building a **HealthConnect Data Intelligence** using **SQL, PLSQL**, aimed at health care data from multiple source files into a centralized persistent system. This warehouse will enable better decision-making through the availability of historical data, consolidated reporting, and improved data analytics for healthcare data.

The primary objectives include:

- Streamlining the healthcare data pipeline.
- Enabling **real-time Analytics and Reporting** through SQL & PLSQL.
- Applying **business logic** to maintain **data quality** and consistency across various datasets.

2.0 Scope

The scope of this project includes:

- **Data extraction** from source systems like flat files (.csv)
- **Data loading** into the **SQL Server.**
- **Versioning** of data using **SCD Type 1** for tracking data changes.

3.0 Assumptions and Constraints

Assumptions:

- **Business Rules**: Clear business rules for data validation are defined.
- **Data Quality**: Source systems provide accurate data with minimal errors.
- **Tool Availability**: (SQL, PLSQL, etc.) are pre-configured.
- **SCD Type -1 Support**: Data Changes tracking (SCD1) is supported for healthcare data.
- **Stakeholder Engagement**: Business stakeholders are available for input.

Constraints:

1. **Data Volume**: High volumes of data may strain SQL Server resources.
2. **Real-time Reporting**: Near real-time, not true real-time reporting due to processing delays.
3. **Data Source Delays**: Pipeline depends on timely data from external sources.

4.0 DESIGN CONSIDERATIONS

4.1 Naming Standards

DatabaseName:

[Enviornment_ProjectName_LayerName]

Example : dev_HealthConnect_cleansed

TableName:

[layerName_tablename]

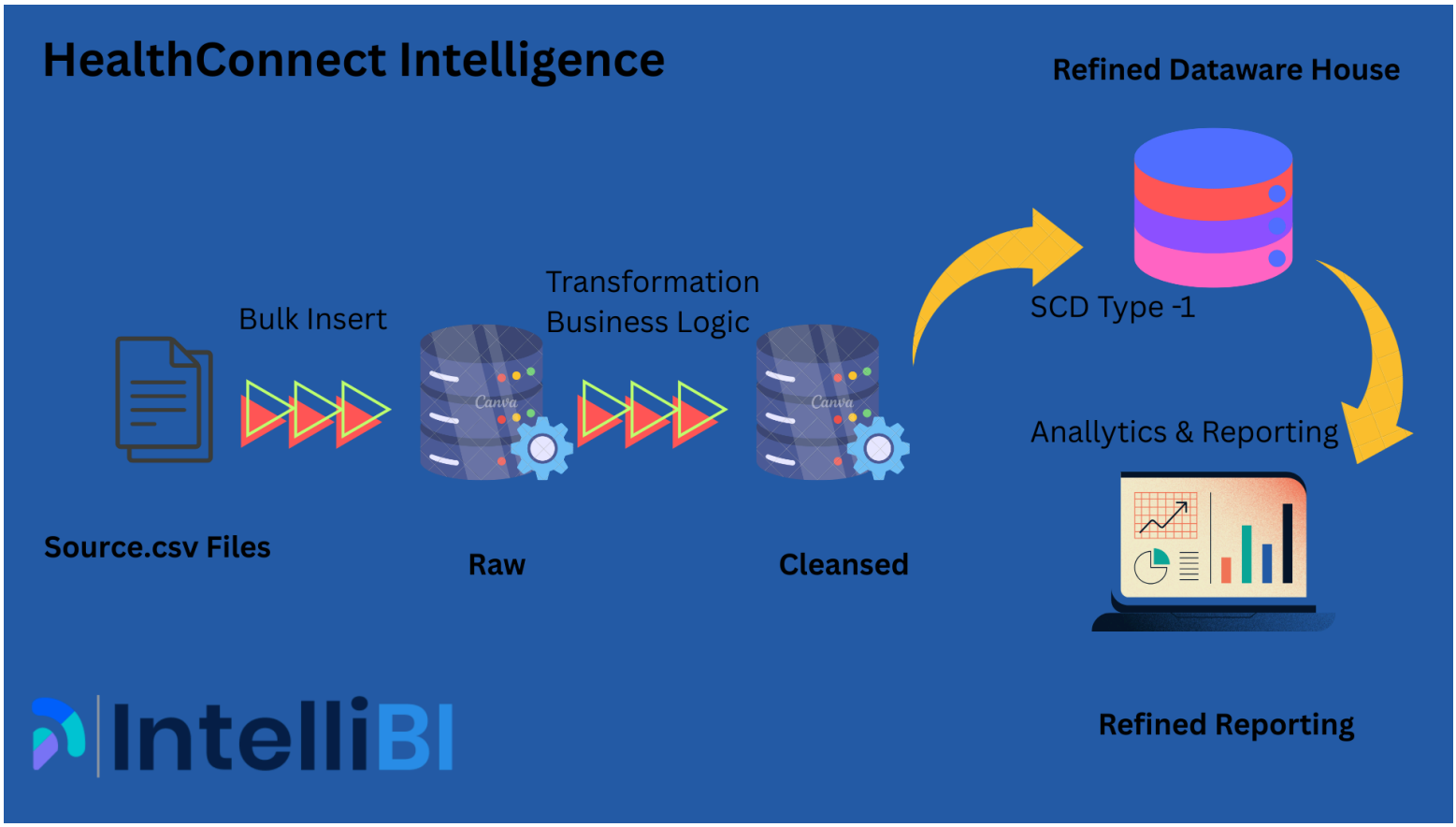
Example : raw_patient

StoreProcedureName:

[Proc_HealthConnect_SourceLayerName_To_TargetLayer_Load]

Example : **Proc_HealthConnect_Raw_To_Cleansed_Load**

5.0 HIGH LEVEL DESIGN (PROJECT ARCHITECTURE DIAGRAM)



Technology components

Technology	Function
SQL	For Extract, Transform and Load
PLSQL	Build PipeLine and Process the data.

Sources:

- Flat **Files (e.g .csv)**: Available in the source folder, these files will be uploaded into Local Server everyday for the processing.

Environments Setup: Development

Create three databases as follows for **dev** Environment:

1. **dev**_HealthConnect_raw
2. **dev**_HealthConnect_cleansed
3. **dev**_HealthConnect_Refined

Create three databases as follows for **qa** Environment:

4. **qa**_HealthConnect_raw
5. **qa**_HealthConnect_cleansed
6. **qa**_HealthConnect_Refined

Create three databases as follows for **prod** Environment:

7. **prod** _HealthConnect_raw
8. **prod** _HealthConnect_cleansed
9. **prod** _HealthConnect_Refined

Note: Set up only in the **DEV** environment for now.

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1. Inbound Layer:

1. Inbound Layer Requirements

1. **Data Loading Convention**
    - **File Naming Convention**:
        - **Directory Name**: D:\HelthConnect\InboundFiles\
        - **File Name**: TableName_<todaystimestamp>.csv
        - **Example**:
            - **Source Folder Path**: D:\HelthConnect\InboundFiles\patient_20250930.csv

2. **Files to Fetch from Source**:

   Source Files Location: Source Files

Files to Load from Inbound Local Storage:

- D:\HelthConnect\InboundFiles\claims_yyyymmdd.csv
- D:\HelthConnect\InboundFiles\diagnoses_yyyymmdd.csv
- D:\HelthConnect\InboundFiles\encounters_yyyymmdd.csv

- D:\HelthConnect\InboundFiles\medications_yyyymmdd.csv
- D:\HelthConnect\InboundFiles\patients_yyyymmdd.csv
- D:\HelthConnect\InboundFiles\payers_yyyymmdd.csv
- D:\HelthConnect\InboundFiles\procedures_yyyymmdd.csv
- D:\HelthConnect\InboundFiles\providers_yyyymmdd.csv

3. **Initial Load / Extract Data**:

   Flat File : Perform a full data extraction from the source system into the RAW layer.
   Load Type: Truncate & Load

   Unit Test Case Documents: LLD_TBL_FILE_PATIENTS_TO_RAW
   Data Model: Raw Layer Data Model

4. **Second Load from Raw to Cleansed Layer Tables:**
   Load the data from the RAW layer (**dev_HealthConnect_raw**) to the **Cleansed** layer with some basic data cleaning and business logic and audit Columns.
   1. Remove leading and trailing spaces for the string datatype columns.
   2. Convert all string datatype columns into upper case letters.
   3. Add **Load_Date** column and load system date into **load_date** column.
   4. **Load Type**: Truncate & Load

   Unit Test Case Documents: LLD_TBL_RAW_PATIENTS_TO_CLEANSED
   Data Model: Cleansed Layer Data Model

5. **Third Load from Cleansed to Refined Layer Tables:**
   Load Type: Slowly Changing Dimension Type 1
   Please refer to the attached Document for this.

   Unit Test Case Documents: LLD_TBL_CLEANSED_PATIENTS_TO_REFINE
   Data Model: Refined Layer Data Model

   SCD-Type-1 Example

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Orchestration For Flat File Source:

1. Source To Raw Pipelines:

1. **Create end-to-end pipeline (Store Procedure) for each .csv File to SQL Sever:**
    - Truncate & Load data from the source file to the SQL Server Raw Layer.
    - Use the parameterization for filename,databasename, schemaname, tablename

2. Master Pipeline (Store Procedure) for SQL server Raw Layer:

- Create an master pipeline (Store Procedure)to:
    - Execute all store procedures involved in Source to Raw data Layer.

2. Raw To Cleansed Pipelines:

1. **Create end-to-end pipeline (Store Procedure) for Raw to Cleansed SQL Sever:**
    - Truncate & Load data from the SQL Server Raw Layer to Cleansed Layer.
    - Use the parameterization for databasename, schemaname

2. Master Pipeline (Store Procedure) for SQL server Raw Layer:

- Create an master pipeline (Store Procedure)to:
    - Execute all store procedures involved from Raw to Cleansed data Layer.

3. Cleansed To Refined Pipelines:

1. **Create end-to-end pipeline (Store Procedure) for Cleansed to Refined SQL Sever:**
    - Load data from the SQL Server Cleansed to Refined Layer.
    - Use the parameterization for,databasename, schemaname

2. Master Pipeline (Store Procedure) for SQL server Raw Layer:

- Create an master pipeline (Store Procedure)to:
  Execute all store procedures involved from Cleansed to Refined data Layer.

3. Capture all Log Details of each Store Procedure.

4. Scheduling:

- Schedule the master pipeline to run daily at 10 PM IST.

4. Refined dw to Reporting :

1. Create Views for each to get the expected reports or KPI's based on Refined_dw Layer.

2. **KPI Requirement Document:** HealthConnectKPI's

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